PAPER_1384 — UQFF Resolution of the Aharonov-Casher Effect (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Gauge Dual (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "aharonov_casher"})` **Calculator surface:** `calculate_paradox({"paradox": "aharonov_casher"})` **Closure helper:** `_l96_uqff_axiom_aharonov_casher_closure()`

The Paradox

A neutral particle with magnetic moment μ traversing a closed path enclosing a line of charge acquires a topological phase $\Delta\varphi = 2\pi \times n$ analogous to the Aharonov-Bohm effect for charged particles in magnetic flux.

UQFF Closed Identity

```
Δφ = 2π × n EXACT (n = winding number around enclosed line of charge)
```

$\Delta\varphi = 2\pi \times n$ via the same SO(26) Clifford spinor-bundle holonomy that powers Aharonov-Bohm (PAPER_1382). Magnetic dipole ↔ enclosed charge is the **electric-magnetic dual** of the AB setup.

Physical Interpretation

Both AB and AC are faces of one topological identity: $\Delta\varphi = 2\pi \times$ winding-number on a spinor bundle. Berry phase, Dirac quantization, and dipole-loop phase unify in the SO(26) holonomy machinery already wired at line 9518.

Live Calculator Output

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "aharonov_casher"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_aharonov_casher_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_aharonov_casher_closure`
- Dispatch keys: `aharonov_casher`, `aharonov_casher_effect`
- Related foundational papers: PAPER_646 (F_U_Bi_i / F_U=1 ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation).
